

# ROYAL FITNESS CLUB

## Body Transformation Blueprint

### MODULE 7

## Lifestyle Optimisation: Sleep, Stress & NEAT

|             |                 |
|-------------|-----------------|
| Duration    | 35 minutes      |
| Course Code | CS_B3           |
| Level       | Beginner        |
| Format      | Study Guide PDF |

Part of the Royal Fitness Club Professional Certification Program.

SECTION 1: Sleep — The Cornerstone of Transformation

Sleep Deprivation: The Hidden Saboteur

Of all lifestyle factors that influence body composition, sleep is the most underrated and the most actionable. The evidence is unambiguous: insufficient sleep systematically undermines every aspect of body transformation.

| Sleep Deprivation Effect                                                             | Mechanism                                                  | Magnitude                                                     |
|--------------------------------------------------------------------------------------|------------------------------------------------------------|---------------------------------------------------------------|
| Increased fat gain (less muscle) from elevated cortisol plus sleeping 6.5h per night | Elevated cortisol plus sleeping 6.5h per night             | Only 18% of weight gained is muscle vs 50% at 8.5h (Nederhof) |
| Increased hunger                                                                     | Ghrelin +15–20%; leptin -40–50%                            | Extra 300–500 kcal consumed per day on sleep-deprived days    |
| Reduced testosterone                                                                 | Restricting to 5h/night for 1 week drops testosterone      | Equal to 10–15 years of aging                                 |
| Impaired training performance                                                        | Reaction time, strength, power output, endurance           | 5–15% performance reduction across measures                   |
| Reduced insulin sensitivity                                                          | Cortisol + reduced GLUT4 expression                        | Equivalent to pre-diabetic state after 6 days of 4h sleep     |
| Impaired muscle protein synthesis                                                    | Reduced GH; increased cortisol; impaired protein synthesis | More protein needed; greater risk of overreaching             |

Sleep Extension — The Performance Drug

If sleep could be bottled as a supplement, it would be the most powerful performance enhancer ever discovered. Sleep extension studies show:

- Stanford Basketball study (Mah 2011): 10h sleep/night → 5% sprint speed, 9% free throw accuracy
- Rugby players: 2 additional hours of sleep/night → 20–24% faster 40m sprint times
- Swimmers: 9–10h sleep vs normal → improved reaction time, turn time, kick stroke
- General training: muscle protein synthesis peaks during N3 deep sleep; 8+ hours required to complete multiple cycles

SECTION 2: Stress Management for Body Composition

The Cortisol-Body Composition Connection

Chronic psychological stress activates the HPA axis, producing chronically elevated cortisol. For body transformation, this is deeply counterproductive:

- Cortisol promotes visceral fat deposition (visceral adipocytes have 4× cortisol receptors vs subcutaneous)
- Cortisol stimulates appetite for high-calorie foods (especially sugar and fat)
- Cortisol suppresses testosterone (antagonist relationship)

- Cortisol inhibits growth hormone secretion (impairs overnight recovery)
- Cortisol is catabolic — promotes muscle protein breakdown for gluconeogenesis

## Evidence-Based Stress Reduction Strategies

| Strategy                                             | Effect on Cortisol                                                  | Time Required        | Evidence Level                        |
|------------------------------------------------------|---------------------------------------------------------------------|----------------------|---------------------------------------|
| Diaphragmatic breathing (Active diaphragm breathing) | Acute cortisol reduction; parasympathetic; acute cortisol reduction | 5-10 min daily       | A — immediate effects well-documented |
| Meditation (mindfulness-based)                       | Reduction in salivary cortisol over 8 weeks                         | 10-20 min daily      | A — strong meta-analytic support      |
| Nature exposure (green/blue spaces)                  | Modest cortisol reduction in 20 min                                 | 20-30 min outdoors   | B — consistent findings               |
| Social connection (positive)                         | Oxytocin release → cortisol reduction                               | Varies               | B — strong epidemiological data       |
| Progressive muscle relaxation                        | Reduces sympathetic activity; lowers cortisol                       | 15-20 min            | B — clinical evidence                 |
| Journaling (Expressive writing)                      | Reduces emotional load; reduces rumination                          | 10-15 min before bed | B — Pennebaker paradigm               |

### SECTION 3: NEAT — Non-Exercise Activity Thermogenesis

## NEAT: The Secret to Sustainable Fat Loss

Non-Exercise Activity Thermogenesis (NEAT) encompasses all calorie-burning movement that isn't deliberate exercise: walking to work, climbing stairs, fidgeting, standing, household chores, grocery shopping, spontaneous movement.

**NEAT varies enormously between people:** Studies show NEAT varies up to 2000 kcal/day between individuals of similar body size. This variation explains why some people seem to "eat anything and not gain weight" — they have high spontaneous NEAT.

| NEAT Activity                | Calories Burned/Hour         | vs Sedentary Sitting                          |
|------------------------------|------------------------------|-----------------------------------------------|
| Standing                     | 80–100 kcal/hr               | vs sitting (70 kcal/hr) = 10–30 kcal/hr extra |
| Walking (slow, 4km/hr)       | 150–200 kcal/hr              | 80–130 kcal/hr extra                          |
| Walking (brisk, 6km/hr)      | 250–350 kcal/hr              | 180–280 kcal/hr extra                         |
| Fidgeting/postural movement  | 50–100 kcal/hr above sitting | Varies widely between individuals             |
| Household chores             | 150–250 kcal/hr              | 80–180 kcal/hr extra                          |
| Standing desk (full workday) | ~100 extra kcal/8hr day      | Small but accumulates                         |

**Why NEAT matters more than gym cardio for most people:** A person who trains 5 hours/week but sits for 12+ hours/day has a fundamentally lower total calorie burn than an "untrained" person who walks everywhere, stands at work, and takes the stairs. Exercise alone cannot compensate for a sedentary lifestyle.

## NEAT Optimisation Strategies

- 10,000 steps/day target: burns 300–500 extra kcal with no recovery cost; walk after meals (improves glucose control)
- Standing desk or desk riser: adds 30–60 min standing per day; reduces sedentary time
- Walk during phone calls: accumulates 30–60 min of movement on work days without time cost
- Stairs > lift: 5 flights of stairs/day = 2km equivalent of NEAT
- Parking deliberately further: 10 min walk each way = 100–150 kcal extra/day
- Active commute: cycling or walking to work transforms commute time into NEAT

**NEAT suppression during dieting:** When calories are reduced, the body unconsciously reduces NEAT by 200–400 kcal/day — less fidgeting, slower movement, more resting. This is a key component of metabolic adaptation. Consciously targeting steps (using a step counter) counteracts this suppression.

### Module 7 (CS\_B3) Key Takeaways

- Sleep is non-negotiable: 5.5h vs 8.5h of sleep changes the muscle:fat ratio of weight gained from 50:50 to 18:82
- Ghrelin rises and leptin falls with sleep deprivation — adding 400–500 kcal of daily hunger
- Chronic cortisol from stress selectively deposits visceral fat and suppresses testosterone and GH
- Meditation, diaphragmatic breathing, and nature exposure are evidence-based cortisol reducers
- NEAT varies up to 2000 kcal/day between similar-sized individuals — the key to "effortless" leanness
- Protect NEAT during dieting by consciously tracking steps (target 8000–10,000/day) as the body suppresses it unconsciously

1. What does the Nedeltcheva 2010 study reveal about sleep duration and body composition changes during a caloric surplus?
2. Explain the three mechanisms by which sleep deprivation increases daily caloric intake.
3. What is NEAT suppression and how does it contribute to metabolic adaptation during dieting?
4. Compare the NEAT calorie burn of a "gym 5x/week but sedentary otherwise" person vs a "no gym but walks everywhere" person.
5. Describe two breathing techniques that reduce acute cortisol and explain the mechanism.